Midlands Fluid Mechanics Meeting 2025

Edgbaston Campus, University of Birmingham, Birmingham, UK

June 4, 2025

It is our pleasure to welcome you all to the University of Birmingham for the third edition of the Midlands Fluid Mechanics Meeting (MFMM 2025). The first and second editions of this conference were successfully held at Aston University and Leicester University in 2023 and 2024, respectively. MFMM aims to bring together academics from across the Midlands who are engaged in research related to fluid mechanics. The region hosts a substantial body of pioneering work in this field, and the meeting is intended to serve as a platform for researchers to share insights from their respective institutions. Equally, it will provide an informal yet productive setting to facilitate the development of new academic collaborations and strengthen regional research networks.

This will be a one-day conference and networking event among the fluid mechanics colleagues in the eight different Universities in the Midlands region. This conference will consist of invited keynote talks from EPSRC and UK Fluids Network representatives, thematic sessions on different fluid mechanics areas, chaired by relevant academics, and a networking lunch. We will organise a panel discussion on how the University of Birmingham can play a key role in establishing the Midlands Fluid Mechanics Group as the regional centre of excellence and contribute to the UK Fluids Network 2.0 bid. We are also planning to organise an exhibit from the regional industrial partners to initiate a collaborative dialogue between the academics and local industry partners.

We thank the College of Engineering and Physical Sciences, University of Birmingham, for the Quality-Related Research (QR) funding, and Birmingham Environment for Academic Research (BEAR), Centre for Optimisation, Simulation and Data Driven Modelling (COSIMO), and the Institute for Data and AI for supporting us in organising the event!

We are looking forward to hosting you all here!
With very best wishes,
MFMM 2025 Local Organising Committee

(Chandan Bose, Dave Smith, Meurig Gallagher & Hibah Saddal)

Local Organising Committee

Dr. Chandan Bose
Assistant Professor of Aerospace Engineering, University of Birmingham
Webpage: **https://www.birmingham.ac.uk/staff/profiles/metallurgy/bose-chandan**

Prof. David J. Smith
Professor of Applied Mathematics, University of Birmingham
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Dr. Meurig Gallagher
Assistant Professor, Institute of Metabolism and Systems Research,
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Ms. Hibah Saddal
PhD Student, Aerospace Engineering, University of Birmingham
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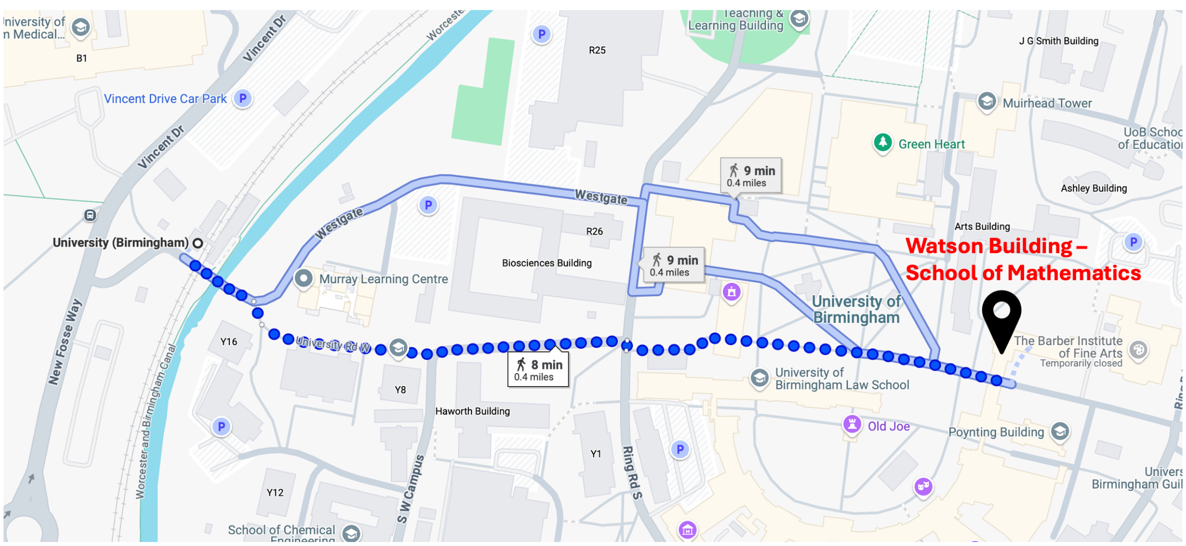
Conference Venue

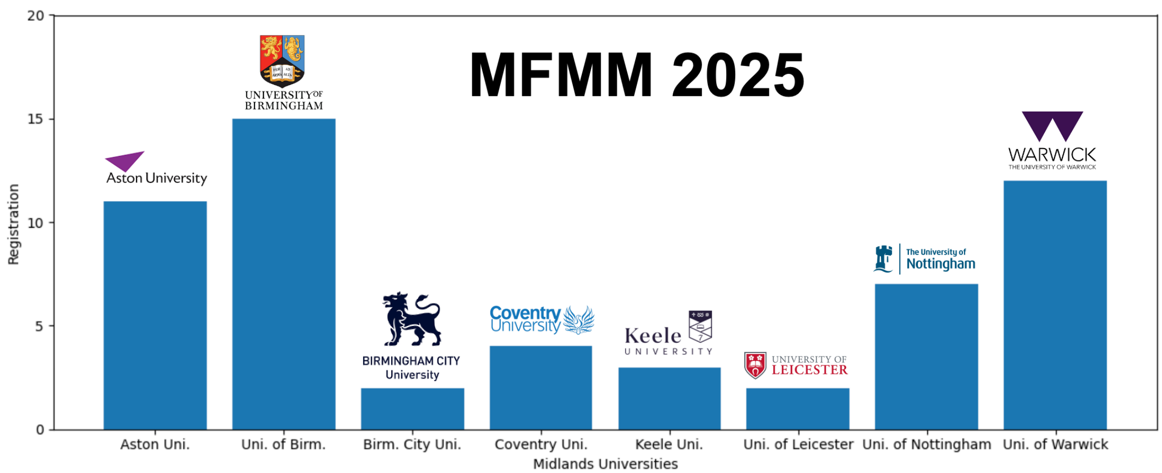
Watson Building (R15), School of Mathematics, University of Birmingham
Conference Talks: Lecture Theatre B
Lunch and Coffee Breaks: B16



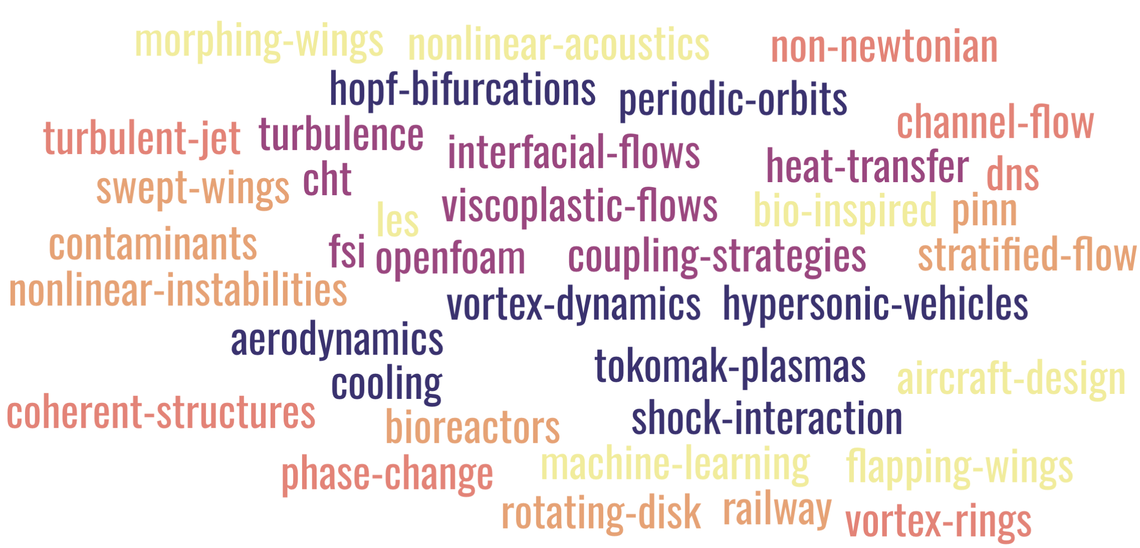
Campus Map: https://www.birmingham.ac.uk/documents/university/edgbaston-campus-map.pdf

Registration Details

Total Registration: 57
Total number of oral presentations: 24



Key Topics Included



Schedule

Timing	Activity
08:30-09:00	Arrival
09:00-09:10	Welcome Address
09:10-09:30	Keynote Lecture
09:30-10:30	Morning Session 1 - Chair: Prof. David Smith
10:30-10:45	Mid-Morning Break
10:45-11:45	Morning Session 2 - Chair: Prof. Peter J. Thomas
11:45-12:30	Panel Discussion on UK Fluids Network 2.0
12:30-14:00	Group Photo and Lunch Break
14:00-15:00	Afternoon Session 3 - Chair: Dr. Paul Griffiths
15:00-15:15	Mid-Afternoon Break
15:15-16:15	Afternoon Session 4 - Chair: Prof. Andrew McMullan
16:15-16:30	Mid-Afternoon Break
16:30-17:20	Afternoon Session 5 - Chair: Dr. Chandan Bose
17:20-17:30	Concluding Remarks

Overview of Talks

Welcome Session	
09:00-09:10	Welcome Address
09:10-09:30	Keynote Address (TBC)
Session 1: CFD 1	**Chair: Prof. David Smith**
09:30-09:42	Dr. Junho Park (Coventry University)
09:42-09:54	Dr. Emad Tandis (Aston University)
09:54-10:06	Ms. Lois Martin & Mr. Ethan Warman (Uni. of Birmingham)
10:06-10:18	Ms. Hibah Saddal (University of Birmingham)
10:18-10:30	Dr. David Adebayo (Aston University)
Session 2: EFD	**Chair: Prof. Peter J. Thomas**
10:45-10:57	Prof. Peter J. Thomas (University of Warwick)
10:57-11:09	Dr. Patrick Geoghegan (Aston University)
11:09-11:21	Ms. Rebecca Lewis (Aston University)
11:21-11:33	Dr. Hassan Hemida (University of Birmingham)
11:33-11:45	Dr. Qian Li (University of Birmingham)
11:45-12:30	**Panel Discussion on UK Fluids Network 2.0**
Session 3: Math. and Data-Driven FM	**Chair: Dr. Paul Griffiths**
14:00-14:12	Dr. Liam Escott (Aston University)
14:12-14:24	Mr. Serkan Ayan (University of Warwick)
14:24-14:36	Dr. Ali Haghiri (University of Leicester)
14:36-14:48	Mr. Freddie Jensen (University of Warwick)
14:48-15:00	Dr. Paul Griffiths (Aston University)
Session 4: Turbulent Flows	**Chair: Prof. Andrew McMullan**
15:15-15:27	Ms. Kate Goodman (Keele University)
15:27-15:39	Mr. Amrit Cassim (University of Birmingham)
15:39-15:51	Mr. Zekun Ma (University of Warwick)
15:51-16:03	Dr. Chris Pringle (Coventry University)
16:03-16:15	Mr. Dao Zhou (University of Birmingham)
Session 5: CFD 2	**Chair: Dr. Chandan Bose**
16:30-16:42	Dr. Mirco Magnini (University of Nottingham)
16:42-16:54	Mr. Arjun Shergil (University of Birmingham)
16:54-17:06	Mr. Daniel Butler (University of Birmingham)
17:06-17:18	Dr. Chandan Bose (University of Birmingham)

Session 1 - CFD 1 - Chair: Prof. David Smith

Talk 1: Dr. Junho Park (*Coventry University*)

Non-linear instabilities of stratified Taylor-Couette flow

J. Park, J. Gianfrani, A. Kumar

Various naturally occurring and engineering flow systems are commonly under multi-physical effects such as stratification (i.e. density or temperature variation along the vertical/gravitational direction) or rotation acting on the horizontal plane. It has been known that the Coriolis force induced by rotation can suppress or promote instability and turbulence depending on the direction and strength, following the Rayleigh's criterion for centrifugal instability (CI). Stable stratification plays a role of suppressing the vertical motion of fluids leading to the delay of the onset of CI and its turbulence. Unlike these known effects, recent research has unveiled new aspects on the role of stratification and rotation; for instance, there is a new type of instability called the strato-rotational instability (SRI) that occurs due to a resonance of inertia-gravity waves (IGWs). SRI does not follow the traditional Rayleigh's criterion and can be promoted in the Rayleigh-stable regime if there is the IGW resonance. Furthermore, the non-linear dynamics of CI and SRI remains yet to be explored further. In the talk, I will present recent numerical simulation results on non-linear instabilities of Taylor-Couette flow, a flow between two concentric cylinders that rotate independently. There will be discussion on the dynamics with various pattern formation that depends on parameters such as the stratification frequency, the Reynolds number, the Prandtl number, etc.

Talk 2: Dr. Emad Tandis (*Aston University*)

Coupling Strategies for Conjugate Heat Transfer Equations in OpenFOAM

E. Tandis

Solving conjugate heat transfer (CHT) problems numerically poses challenges related to accuracy, stability, and computational cost. This study provides a comprehensive analysis of existing OpenFOAM® CHT solvers, discussing the different forms of governing equations and coupling methods. Two enhanced CHT approaches are then introduced: (i) a new partitioned algorithm with improved efficiency, and (ii) a modified version of the monolithic CHT solver from foam-extend-4.0, extended to support multiple regions. The performance of these solvers is evaluated across various transient test cases with different coupling strengths. Results demonstrate the superiority of the proposed monolithic approach over partitioned methods, particularly for cases involving more than two regions. Additionally, the new partitioned approach proves to be more computationally efficient than existing partitioned solvers.

Talk 3: Ms. Lois Martin & Mr. Ethan Warman (*University of Birmingham*)

Wake Dynamics of Pitching Forward and Backward Swept Wings at Low Reynolds Numbers

L. Martin, E. Warman, C. Bose

A clear understanding of aerodynamic characteristics is crucial for improving the design for next generation micro aerial vehicles (MAVs). There has been limited computational effort in comparing the effect of increasing sweep angle (Λ) on backward-swept wings (BSWs) and forward-swept wings (FSWs) at varying Strouhal numbers and Aspect Ratios, which this study aims to address. The open-source code library, OpenFOAM is used to investigate three-dimensional pitching wings. Sweep angles of: $\Lambda = 30, 45, 60$, semi-aspect ratios (sAR) of 2 and 4, and Strouhal numbers (St) of 0.159 and 0.318 are considered to examine the wake dynamics. A chord-based Reynolds number (Re_c) of 1000 is used to observe the separation characteristics of the wing with a prescribed reduced frequency (k) of 1 oscillating between angles of attack (α): $-25°$ to $25°$. For both BSWs and FSWs, the leading-edge vortex (LEV) dominated lift production; the strength of the LEV increased with increased at the cost of coherence across the wingspan, leading to earlier onset of tip stall. Uniquely, the $\Lambda = 45^o$ was the only thrust-generating configuration due to the dynamic stall vortex (DSV) being displaced by the pitching aerofoil, creating a reverse Von Karman vortex street. For FSWs, an increase in aspect ratio was associated with a higher peak coefficient of lift (C_L). The increase in forward sweep resulted in a delayed stall compared to that of the BSWs. The present findings inform the efficient design of futuristic bio-inspired MAV wings.

Talk 4: Ms. Hibah Saddal (*University of Birmingham*)

Wake Dynamics of Bio-Inspired Flaps and Morphing Foils

H. Saddal, C. Bose

This study investigates the wake dynamics of bio-inspired flexible flaps and morphing foils [1] in the low Reynolds number regime, focusing on their effect on the overall aerodynamic performance. The incompressible fluid flow is governed by the Navier-Stokes equations and solved using the finite volume method in OpenFOAM. The structural part is simulated using the finite element based solver CalculiX. A two-way coupling between the fluid and structural solvers is achieved through preCICE, enabling efficient data exchange between both solvers. The radial basis function interpolation technique is required for data mapping at the fluid-structure interface, while the Interface Quasi-Newton Inverse Least Squares (IQN-ILS) acceleration scheme ensures strong coupling. This study leverages these advanced numerical techniques to explore the effects of governing parameters such as dimensionless bending rigidity and mass ratio/density ratio. By varying the flexibility of covert-inspired flaps and optimising their locations, the stall can be significantly delayed, leading to increased lift. Additionally, from morphing wing studies, we also concluded that passive morphing is shown to mitigate spatial-temporal gust effects, improving aerodynamic efficiency. These findings contribute to the optimisation of micro air vehicles and unmanned aerial vehicles, enhancing their flight performance in challenging conditions such as gusty winds and complex manoeuvres. For the two cases mentioned, the underlying physics of vortex-dominated flow is investigated in detail.

1. Li, D., Zhao, S., Da Ronch, A., Xiang, J., Drofelnik, J., Li, Y., De Breuker, R. (2018). A review of modelling and analysis of morphing wings. Progress in Aerospace Sciences, 100, 46-62.

Talk 5: Dr. David Adebayo (*Aston University*)

Investigation of Transpiration Cooling in Hypersonic Vehicles

D. Adebayo[1], R. Lawal[2]
[1] School of Engineering and Technology, College of Engineering and Physical Sciences, Aston University, Aston Triangle, Birmingham B4 7ET, United Kingdom.
[2] Faculty of Engineering and Technology, Centre for Engineering Innovation and Research, University of Wolverhampton, Telford Innovation Campus, Telford TF2 9NT, United Kingdom.

Studies have shown that Thermal Protection System (TPS) designs are critical for hypersonic vehicles due to the large aerodynamic heating at high Mach numbers. The intense heat flux incident upon the leading edges of hypersonic vehicles requires creative thermal management strategies to prevent damage to leading-edge components. Conventional passive Thermal Protection Systems (TPSs) have proven to be effective in objects re-entering the earth's atmosphere, however, the performance of conventional transpiration cooling scheme is weakened by shock interaction.

This study explores an active cooling system embedded into the body of a hypersonic vehicle to cool the area directly close to the opening of the pore. The focus is to compare the cooling performance of a simple coolant pore with a Triply Porous Minimal Structure (TPMS) coolant pore. Computational Fluid Dynamics (CFD) technique was used to conduct simulations at various flow conditions ranging from coolant flow speeds of 5 m/s – 50 m/s, and Mach numbers 0.5 and 5, to identify flow behaviour, coolant response, and potential in each case.

The initial result of this study shows that there is a potential for TPMS cooling pores to effectively cool regions of utmost importance when a proper cooling condition is imposed, without necessarily altering the overall dynamics of the flow conditions or creating secondary flow conditions of its own. It is believed that a combination of several TPS systems can be utilised on hypersonic vehicles depending on the coolant functionality required. By employing the positive characteristics of individual cooling methods at appropriate locations where they may best function to effect cooling might offer a better efficiency.

Session 2 - EFD - Chair: Prof. Peter J. Thomas

Talk 1: Prof. Peter J. Thomas (*University of Warwick*)

Experiments on Vortex Rings in a Rotating Fluid: Decay, Inertial-Wave Field and Spatial Energy Redistribution

P. J. Thomas, O. C. Jackson

Experimental results of a study investigating the dynamics of vortex rings in a water-filled, rotating tank are presented. The vortex rings propagate along the axis of rotation. Coriolis forces arising as a consequence of the background rotation fundamentally alter the dynamics of the rings in comparison to their non-rotating counterparts. Coriolis effects lead to a strong destabilisation of the vortex rings. Rings in non-rotating systems can travel distances of the order of hundred times their ring diameter, relative to their origin, before they have fully lost their structural identity. For rings subject to sufficient levels of background rotation the decay takes place violently within a distance of one ring diameter from their origin. Vortex rings in a rotating fluid have to be anticipated to be accompanied by an inertial-wave field in the fluid surrounding the rings. However, the existence of this inertial wave field had previously never been verified experimentally. Our experimental data demonstrate its existence for the first time by showing that the data begin to increasingly align with the theoretical dispersion curve for inertial waves as the level of the background rotation is successively increased. For sufficiently high levels of rotation there is a very good agreement between the experimental data and the theoretical dispersion curve. We then briefly discuss how the inertial-wave field leads to a fundamental redistribution of the kinetic energy associated with the vortex rings.

Talk 2: Dr. Patrick Geoghegan (*Aston University*)

Title: Challenges of experimental validation of CFD to gain regulatory approval for medical devices

P. Geoghegan [1], J. Lowe [1], E. Barton [1], A. Fratini [1], W. H. Ho [2], J. Simms [1], J. B.. Souppez [1]
[1] Aston University, Birmingham, England
[2] University of Cape town, Cape town, South Africa

A fundamental challenge when using computational fluid dynamics in the development of medical devices for arterial diseases, when obtaining approval from bodies like the Medicines and Healthcare products Regulatory Agency, is meeting the stringent requirements set out by frameworks like ASME Verification and Validation (VV)40. VV40 is a risk-informed credibility assessment framework outlining a process for making risk-informed determinations as to whether a computational simulation is credible for decision-making for a specified context of use. Developing an experimental platform achieving this is challenged by the complex nature of the cardiovascular system. The fluid-structure interaction between blood and the artery presents two key problems. Firstly, blood is a shear-thinning fluid with a non-Newtonian regime at low shear rates, also exhibiting viscoelastic, viscoplastic and time-dependent thixotropic effects. Secondly, compliance of the artery and the resistance of the surrounding material are not easily quantified. Both influence key markers of arterial health: wall shear stress and shear effects on blood cells. Current experimental blood analogues in literature do not account for variation in constituent materials due to thermal effects and even between batches produced by the material manufacturers. Furthermore, there is minimal explanation of management of external arterial resistance presented in literature. This work will present a full rheological characterisation of a blood analogue, mapped to a power law model for non-Newtonian fluids. Physiologically realistic arterial expansion of phantoms to mimic compliance, will be modelled using pressure - area relationship equations. It is anticipated these findings will support future experimentalists in developing reliable experimental platforms for the verification and validation of computational models for biomedical fluid dynamics.

Talk 3: Ms. Rebecca Lewis (*Aston University*)

Towards Experimental Fluid Dynamics of Stirred-Tank Bioreactors

R. Lewis, E. Ross, E. Theodosiou, P. Geoghegan, J.B. Souppez

There is a continuously rising demand for cell culture products, including mesenchymal stem cells (MSCs), due to their many applications in regenerative medicine [1] and cultivated meat [2]. Unfortunately, traditional cell culture methods are inefficient [3], particularly at the scale needed to meet demand. Stirred-tank bioreactors (STBRs) are a more efficient alternative which use microcarriers to increase the surface area-to-volume ratio of the culture [3], providing more space for adherent cells to grow per volume of media. An impeller stirs the culture to increase homogeneity and suspend the microcarriers [4]. Suspension aids attachment by increasing the contacts between cells and microcarriers and allowing cells to use the entire microcarrier surface area [5]. It also improves the cells' access to their nutrient-containing media [6]. However, a balance is required as higher stirring speeds increase turbulence, consequently reducing attachment [5] and placing fluid dynamic stresses like shear stress on the cells [7]. This stress can affect the number or characteristics of the yielded cells [6], potentially reducing their usefulness. This project aims to characterise the fluid dynamics within STBRs of varying sizes to aid the scale-up of cell culture. Particle image velocimetry (PIV) will be used to quantify the velocity field and shear stress within models of 0.1, 1 and 3 L STBRs with varying impeller speeds, heights and geometries. Iterative design improvements will be made to enhance areas of poor flow quality. Together with data from MSC culture, the PIV results will help to determine and explain the optimal flow conditions for MSCs in STBRs.

[1] López-Fernández, Cytotherapy, 26(5), (2024).
[2] Hanga, Biotechnology and Bioengineering, 117(10), (2020).
[3] Wang, Stem Cell Research Therapy, 14(1), (2023).
[4] Bellani, Frontiers in Nutrition, 7, (2020).
[5] Hewitt, Biotechnology Letters, 33, (2011).
[6] Heathman, Biochemical Engineering Journal, 136, (2018).
[7] Ismadi, Processes, 2(4), (2014).

Talk 4: Dr. Hassan Hemida (*University of Birmingham*)

Efficient and Cost-Effective Methods for Controlling Wind-Blown Contaminants on Railway Tracks

H. Hemida, J. Romero, D. Soper, M. Jesson

Railway systems face operational challenges due to wind-blown contaminants, including sand and debris. The aim of this project is to evaluate and optimise various mitigation measures with special emphasis on Sand Mitigation Measure (SMMs) to enhance railway safety, efficiency, and cost-effectiveness. The study focused on assessing existing methods, improving aerodynamic efficiency, and developing hybrid mitigation approaches. Wind tunnel tests were performed at the University of Birmingham using a 9-meter-long test section, while CFD simulations were conducted and validated using the experimental data. Full-scale simulations analysed various debris sizes and densities to evaluate their deposition on railway tracks under different wind conditions.

The Key Findings of this study are:
1. Effectiveness of SMMs It was found that all tested SMMs significantly outperformed the baseline case, demonstrating their ability to trap wind-blown contaminants. The Inclined Wall (SMM-IW) achieved the highest mitigation efficiency (99.95%), followed by other wall-based structures and berms. Ditches (SMM-D) effectively captured larger debris, such as gravel and wood, but struggled with finer particles. None of the SMMs fully prevented contamination from micro-sized airborne particles, indicating a need for alternative mitigation strategies.
2. Impact of Wind Yaw Angle Results showed that increasing the crosswind angle generally improved debris containment across most SMMs. Ditches (SMM-D) performed better at higher wind angles due to enhanced debris retention. The zero-wind condition (0°) was identified as the worst-case scenario, facilitating maximum wind-driven contaminant transport.
3. Cost and Maintenance Considerations It has been found that wall-based structures, while effective, have high initial construction costs. Berms and ditches offer lower-cost alternatives but require varying levels of maintenance. Ditches necessitate frequent clearing, whereas berms require stabilization through vegetation management.

Overall, the findings provide a foundation for railway operators to implement optimised sand mitigation solutions, ensuring sustainable and cost-effective railway operations in high-risk areas.

Talk 5: Dr. Qian Li (*University of Birmingham*)

STORMS: Strategies and Tools for Resilience of Buried Infrastructure to Meteorological Shocks

Q. Li

Buried infrastructure networks, comprising of cables and pipes that transport critical resources like water, gas, electricity, and telecommunication are vital for the functioning of society, especially in urban environments. However, these utility networks are often vulnerable to meteorological shocks, or extreme weather events, such as extreme precipitation which causes floods and droughts, and extreme temperatures. The research aims to develop a comprehensive process for assessing risks to buried infrastructure due to climate changes. It begins with three key inputs: rainfall scenarios, soil properties, and soil moisture scenarios. Rainfall scenarios represent varying intensities and patterns of precipitation that might lead to surface water-related risks. Soil properties, including characteristics like grain size and soil depth, determine how the soil responds to water infiltration and mechanical stress. Soil moisture scenarios, derived from climate models, project future conditions of soil saturation under changing climatic patterns. These inputs feed into two main categories of risk. The first category is the risk from surface water, particularly focusing on soil erosion and accumulation. Heavy rainfall can erode the soil surrounding buried assets, such as pipelines, potentially exposing them and compromising their stability. Soil accumulation can also lead to increasing pressure to underground pipelines, potentially damaging infrastructures. The second category is the risk from ground movement, which is divided into two subtypes. One is the loss of load-bearing capacity, where water-saturated or weakened soil loses its ability to support the infrastructure, causing potential subsidence or structural failure. The other subtype is differential subsidence or expansion, where uneven changes in soil moisture or properties create localised ground movement, resulting in stress and deformation of the infrastructure. The resulting framework supports robust climate resilience planning by enabling targeted risk evaluation of underground infrastructure systems.

Session 3 - Mathematical and Data-Driven Fluid Mechanics - Chair: Dr. Paul Griffiths

Talk 1: Dr. Liam Escott (*Aston University*)

Improving stability of a Newtonian boundary layer via non-Newtonian injection

L. Escott, P. Griffiths

The study of boundary layer flows involving Newtonian fluids, with or without suction or injection, has been a subject of interest for researchers in fluid dynamics for more than a century. The aim of our study is to extend traditional boundary layer-type analyses to consider a natural phenomenon that has been observed in some species of fish. Studies have shown that certain fish secrete a small amount of fluid with complex properties, which has the potential to reduce their drag in water. Our research aims to model the behaviour exhibited by these fish via a modified boundary layer analysis.

Given a two-tier system of immiscible fluid layers, we investigate the flow profiles for an injection of shear-thinning fluid. We consider a variety of parameters within the Carreau-Yasuda model, along with injection profiles, to evaluate the linear stability characteristics. Two modes of instability are determined, related to the viscosity stratification within the flow and interfacial contributions respectively. An analysis of the critical Reynolds number, at which we anticipate the transition to turbulence, is provided. Under the correct choice of shear-thinning parameters, related to the injected fluid, our analysis shows that linear instability can be delayed significantly compared to the no-injection case study. Further, we present an energy analysis of component terms and investigate the propagation of perturbation quantities in the spanwise direction.

Talk 2: Mr. Serkan Ayan (*University of Warwick*)

Approximation of Rough Rotating-Disk Boundary Layers via Physics-Informed Neural Networks

S. Ayan[1], P. J. Thomas[2], S. G. Garrett[3], B. Alveroğlu[4], P. T. Griffiths[5]

[1]Visiting PhD Student at University of Warwick & PhD Student at Bursa Technical University
[2]Fluid Dynamics Research Centre, School of Engineering, University of Warwick
[3]School of Engineering & Innovation, Aston University
[4]Department of Mathematics, Bursa Technical University
[5]School of Engineering & Applied Science, Aston University

In this talk, the flow problem on a rough rotating disk will be discussed using Physics Informed Neural Networks (PINNs). PINNs are a specialised deep learning model that provides an alternative method for solving initial/boundary value problems. In this way, it generates a residual network containing differential equations and incorporates it into the loss function. Therefore, such a PINN model has been trained by incorporating the boundary value problem -dimensionless Navier Stokes equations, continuity equation and boundary conditions- into the loss function. The model training was performed at 1000 collocation points selected in the boundary layer and observation points based on numerically verified observation data obtained from a fixed radial position were also included to improve solution accuracy. Upon convergence, one thousand laminar velocity profiles—each corresponding to a selected boundary-layer training point—were generated separately for isotropic and anisotropic roughness cases. The mean flow profiles that were obtained by the numerical techniques, extracted independently of the radial coordinate, exhibited close agreement with the PINN predictions. For accuracy assessment, numerical simulations and PINNs outputs obtained in the MW approach over a range of roughness coefficients are compared.

Talk 3: Dr. Ali Haghiri (*University of Leicester*)

Accelerating Aircraft Design: Machine Learning for Predicting Aerothermal Performance

A. Haghiri, R. D. Sandberg

Enhancing aircraft efficiency and performance is a central goal in advancing sustainable aviation and future transport systems. Machine learning (ML) offers powerful tools to accelerate design processes and improve predictive accuracy in aerodynamic and thermal analyses. This study explores how ML can effectively bridge high-fidelity simulations and low-order models, enabling faster, more reliable evaluations during early design stages.

This study presents a data-driven modelling framework using gene expression programming (GEP) to improve turbulent heat flux predictions in Reynolds-averaged Navier–Stokes (RANS) calculations. The models are trained on large eddy simulation (LES) data from nine wall jet configurations with coflow, covering diverse flow and geometric conditions. By targeting a variable turbulent Prandtl number derived from velocity and temperature gradients, the developed closures demonstrated strong performance and generalisation.

A posteriori testing showed significant improvement in adiabatic wall effectiveness predictions across all cases—even those not seen during training. The results highlight both the potential and limitations of ML-based closures, particularly in flows influenced by strong unsteadiness, where improvements to the RANS velocity field may also be required.

This work underscores the value of integrating ML with physics-based modelling to develop accurate, application-specific tools for complex aerodynamic and thermal systems, with broad implications for cleaner, more efficient aircraft design.

Talk 4: Mr. Freddie Jensen (*University of Warwick*)

Observations from modelling nonlinear acoustics in brass instruments

F. Jensen

We discuss interesting features of a model combining weak nonlinearity and complex geometry in duct acoustics without flow, with applications to sound in brass instruments. Topics discussed here include curvature-induced plane-wave tunnelling, a method of quantifying the speed of sound around bends, an ambiguity around forward/backward decomposition, and a new test case in the study of nodes and turning points.

Talk 5: Dr. Paul Griffiths (*Aston University*)

Modelling viscoplastic interfacial flows in the low Reynolds number regime

P. T. Griffiths[1], D. Xu[2], L. Davoust[3]

[1] School of Engineering & Innovation, Aston University, Birmingham B4 7ET, UK

[2] Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong, PR China

[3] Univ. Grenoble-Alpes, CNRS, Grenoble-INP (Institute of Engineering Univ. Grenoble Alpes), SIMAP, F-38000 Grenoble, France

Molten aluminium alloys, widely used for the casting of lightweight parts, tend to oxidise very quickly when first exposed to the atmosphere at ambient conditions. Thin oxide films, usually less than one micrometer thick, develop at the liquid metal-air interface which, in turn, affects casting processes of, for instance, aluminium (Patouillet et al. (2021)). The oxide film can be encapsulated and further dragged away into the bulk under the effect of surface agitation (Divandari & Campbell (2004)). Campbell (2006) demonstrated that the encapsulation process necessarily leads to embedded oxide films (bi-films) between which air can be trapped. As a result, micro-porosities are formed after the solidification process, which can have major repercussions on the quality and fatigue life of the cast parts.

As a consequence of the oxide film, the selection of the right boundary condition to be applied along the surface of the liquid metal flow is not straightforward because of the complexity of the stress balance between the oxidised molten metal and the ambient surroundings (air). Irrespective of the mechanical behaviour of the oxide layer that covers the surface, one of two boundary conditions are usually considered by default: either a slip condition as usual for water-based flows, or a no-slip condition when the oxide film is considered to behave as a deformable wall. In fact, the reality is somewhat in between these two conditions (Patouillet & Davoust (2021)): the continuous formation of an oxide film at the metal-air interface leads to a local and evolutive change in the boundary condition that can strongly affect the surface flow by increasing viscous dissipation.

Recent experimental investigations by Delacroix et al. (2018) have shed new light on the complex dynamics associated with these liquid-metal-type flow problems. Both the non-Newtonian behaviour of the oxide layer over the melted metal surface and the curvature of the interface due to wetting effects were observed via scanning electron microscopy techniques. In this study we develop a mathematical model to describe flows of this nature. Working in the low Reynolds number regime, we make use of the Bingham fluid model in order to capture viscoplastic surface effects.

In this study we develop both asymptotic and numerical analyses and report on the surface characteristics (velocity and shear profiles) as well as the important effects of surface curvature.

Patouillet et al., AIChE J., 67 (7), 17230 (2021)

Divandari & Campbell, Int. J. Cast Met. Res., 17 (3), 182-187 (2004)

Campbell, Mater. Sci. Technol., 22 (2), 127-145 (2006)
Patouillet & Davoust, Chem. Eng. Sci., 231, 116328 (2021)
Delacroix et al., Rev. Sci. Instrum., 89 (1), 015103 (2018)

Session 4 - Turbulent Flows - Chair: Prof. Andrew McMullan

Talk 1: Ms. Kate Goodman (*Keele University*)

Hydrodynamic simulations of turbulent convection in stellar interiors

K. Goodman, R. Hirschi, V. Varma

The evolution and fate of massive stars are governed by complex, multiscale processes occurring deep within their interiors. Among these, turbulent convection and mixing across convective boundaries play a central role in the redistribution of energy and chemical species, influencing whether a star explodes as a supernova or collapses into a black hole.

Traditional one-dimensional models of massive stars approximate these effects over the whole lifetime of the star, using simplified diffusion schemes which cannot capture the dynamic behaviour of convective boundaries. To address this, we use PROMPI, an Implicit Large Eddy Simulation (ILES) code, to model compressible, stratified convection with realistic nuclear energy generation. Our three-dimensional simulations of convective shells within massive stars explore convective boundary mixing (CBM), showing that turbulent entrainment scales with the bulk Richardson number.

Due to the computational demands of these high-resolution, high Reynolds number flows, we can only simulate thousands of seconds compared to the millions of years that massive stars live. To bridge this gap, results from our 3D simulations are used to inform and improve the accuracy of 1D stellar models that can evolve massive stars over their full lifetimes. In particular, we use our results to enhance the treatment of convective mixing in 1D, allowing us to extend insights from turbulent flow regimes to the stellar models that can evolve until collapse.

Our work highlights the relevance of astrophysical flows for studying buoyancy-driven turbulence, turbulent entrainment and mixing in compressible, stratified systems at high Reynolds number. By studying these extreme regimes, we contribute to a broader understanding of turbulence, boundary layer dynamics, and mixing in complex flow environments.

Talk 2: Mr. Amrit Cassim (*University of Birmingham*)

Coherent structures in turbulent jets from round and serrated nozzles

A. Cassim, Z. N. Wang

Prior studies have shown that serrated nozzles are capable of reducing low-frequency jet noise ($St < 1$) compared to round nozzles, with a maximum reduction of about 5 dB observed at aft-angles. However, chevrons increase high frequency noise ($St > 1$), especially at sideline angles. This results in a reduction in overall SPL at aft angles but an increase in overall SPL at sideline angles, with crossover at an inlet angle of roughly 135°. The reduction of low-frequency aft-angle noise has been linked to disruption of azimuthally coherent structures in the turbulent flow field from the serrated nozzle, since it is these structures that are primarily responsible for this type of noise. The increase of high-frequency sideline noise has been linked to an increase in fine-scale turbulence caused by enhanced mixing due to the chevron-induced streamwise vortices.

Few studies to date have used data-driven methods to extract coherent structures from the flow-field of a jet issuing from a serrated nozzle. In this presentation, we show results obtained from performing Spectral Proper Orthogonal Decomposition (SPOD) on LES data for a Mach 0.7 compressible serrated jet with 16 chevrons. The way in which the chevrons modify and redistribute energy between Kelvin-Helmholtz (KH) and Orr wavepackets as well as streaks will be demonstrated. Chevrons decrease the relative energy of KH wavepackets close to the nozzle while increasing the relative energy of streaks, with this effect being most pronounced as St approaches 0. In addition, the chevrons modify the near-field optimal SPOD spectrum in a similar way to the far-field acoustics. The link between the near and far field is investigated further using acoustic matching calculations through the use of Lighthill's acoustic analogy.

Talk 3: Mr. Zekun Ma (*University of Warwick*)

DNS of a temporally accelerating turbulent channel flow

M. Zekun, Y. M. Chung

Understanding turbulence in accelerating channel flows is essential both for advancing fundamental turbulence theory and for predicting transport phenomena in engineering applications. A notable phenomenon in linearly accelerating channel flows is the significant delay in the turbulence response, characterised by: (i) a delay in the generation of turbulence; (ii) a delay in the redistribution of turbulent energy among the three velocity components; and (iii) a delay in the wall-normal propagation of newly generated turbulence. Previous experiments and direct numerical simulations (DNS) have identified four distinct stages during linear acceleration, with the generation and propagation of new turbulence as dominant features.

However, the mechanisms underlying turbulence generation, particularly the exchange of momentum between near-wall turbulence and the quiescent core region, remain unclear. In this study, we perform DNS to investigate the behaviour of the innermost uniform momentum zone (UMZ) and its relationship to vortical structures during linear acceleration in channel flow.

Talk 4: Dr. Chris Pringle (*Coventry University*)

Using periodic orbits to explain the onset of turbulence in tokamaks

C. Pringle, O. Smith, B. McMillan

As with neutral fluids, plasmas can exhibit subcritical transitions to turbulence and do so in regimes relevant to tokomak fusion. In much of the plasma literature, this turbulence is described in terms of a stochastic collection of linear waves. We instead take ideas from classical shear flows to describe the dynamics in terms of underlying exact (periodic) solutions. For our case of tokomak plasmas, turbulence can be completely suppressed by introducing a background shear flow, whose amplitude is an important control parameter. As this parameter is decreased below a critical value, radially localised structures appear, becoming larger and more complex. We analyse both gyrokinetic simulations of plasma and a simpler fluid model. For the fluid model, we directly solve for a particular class of nonlinear solutions, relative periodic orbits, and determine their stability, thus explaining why these isolated structures appear in initial-value simulations. The increase of complexity as the flow shear is reduced is explained by a series of Hopf bifurcations of these nonlinear solutions, which we quantify via stability analysis. In gyrokinetic simulations, we are able to indirectly determine the underlying relative periodic orbits by imposing appropriate symmetry conditions.

Talk 5: Mr. Dao Zhou (*University of Birmingham*)

Data-driven modelling of coherent structures in a mixing layer

D. Zhou, Z. N. Wang

As a frequent form of pollution, noise can cause both auditory and non-auditory health issues. Turbulent shear flow is a major source that is responsible for the noise of land and air vehicles. Growing evidence suggests that coherent structures within turbulent shear flows are highly correlated to the generation of sound. Linear models are used to predict the coherent structures as instability waves, but tend to underestimate the noise generation, as nonlinear dynamics of coherent structures, such as intermittency, can play a key role in radiating sound. The research in this paper aims to model the nonlinear dynamics of coherent structures in a mixing layer and identify their roles in noise production. The Sparse Identification of Nonlinear Dynamics (SINDy) is used to derive the physics-constrained reduced-order model (ROM) from LES data to capture the key nonlinear dynamics in the low-dimensional basis of Proper Orthogonal Decomposition (POD). An improved eddy viscosity closure is proposed to consider the energy transfer between the resolved and truncated modes.

<u>Session 5 - CFD 2 - Chair: Dr. Chandan Bose</u>

Talk 1: Dr. Mirco Magnini (*University of Nottingham*)

boilingFOAM - an OpenFOAM solver for interface-resolved simulations of liquid-vapour flows with phase change

M. Magnini

Flows with phase change, e.g. boiling, condensation, evaporation, are pervasive in nature and widely utilised in science and engineering for a number of technological processes and for thermal management. The computational modelling of such flows is typically challenging owing to complex evolving two-phase interfaces, strong effects of surface tension, phase change, and coupling with solid regions. In this talk, I will present boilingFOAM, an OpenFOAM solver that we have developed at Nottingham that features interface capturing via a Volume Of Fluid Method, surface tension reconstruction, a phase change model for liquid-vapour evaporation/condensation, conjugate heat transfer between fluid/solid regions, adaptive mesh refinement and processor load balance. The solver has been widely validated against traditional benchmarks such as 1D Stefan and sucking interface problems, 2D/3D spherical bubble growth in an infinite liquid, and experimental measurements for both flow and pool boiling in micron-sized and mm-sized channels. The solver has also been extensively used in national HPC such as ARCHER2 and Midlands+ Sulis. Furthermore, in a recent ARCHER2-eCSE project the solver was coupled with a Molecular Dynamics solver to perform hybrid CFD-MD simulations of boiling, and the results obtained will also be described in this talk. The solver is available open-access in GitHub and we welcome users to utilise it and further develop it.

Talk 2: Mr. Daniel Butler (*University of Birmingham*)

Fluid-Structure Interaction of an impulsively started, flexible NACA 0012 aerofoil

D. Butler, H. Saddal, C. Bose

Flexible aerofoils are fundamental to bio-inspired flight, where structural compliance interacts with unsteady vortex dynamics to enhance aerodynamic performance. This study investigates the fluid-structure interaction (FSI) of an impulsively started flexible NACA 0012 aerofoil, focusing on the role of flexural rigidity and pitch rate in vortex evolution and aerodynamic force generation. Understanding how flexibility and motion timing alter unsteady aerodynamic responses presents a key technical challenge, with significant implications for aeroelastic design. High-fidelity computational simulations were conducted using OpenFOAM for fluid dynamics and CalculiX for structural analysis, coupled through preCICE. Simulations varied the nondimensional flexural rigidity (K_B) and pitch duration to capture a range of unsteady flow phenomena. The results indicate that lower flexural rigidity increases wake asymmetry and flow instabilities, while greater stiffness promotes more organised vortex shedding. Faster pitching amplifies unsteady aerodynamic forces, whereas slower pitching results in smoother transitions towards steady-state conditions. The findings offer new insights into how structural compliance can modulate aerodynamic loads, drawing parallels to natural flyers. These outcomes are expected to inform the design of flexible-winged micro aerial vehicles (MAVs), where balancing agility and stability is critical for operational performance.

Talk 3: Mr. Arjun Shergill (*University of Birmingham*)

Optimising Wake Dynamics and Propulsive Performance in Tandem Pitching Airfoils

A. Shergill, T. Lavender, C. Bose

Bio-inspired propulsion systems modelled after natural flyers and swimmers have driven interest in tandem pitching wing configurations for Autonomous Underwater Vehicles (AUVs) and Micro Air Vehicles (MAVs). These systems aim to exploit wake-foil interactions for enhanced efficiency and thrust. However, it is yet to be clearly understood how varying flapping parameters, particularly Strouhal number and foil spacing, affect wake dynamics and aerodynamic performance. This study builds upon previous research by analysing a range of configurations with varying parameters, including Strouhal number, inter-foil distance, and pitching pivot point, using the open-source library OpenFOAM. It examines how variations in these parameters influence vortex formation and interaction, thrust production, and efficiency. Results indicate that when the hind foil operates at a Strouhal number of 0.5, the mean thrust value of the hind foil reaches a peak, while its efficiency peaks at around 10% at a Strouhal number of 0.3. Furthermore, a foil spacing of two chord lengths is shown to maximise constructive wake and results in larger vortex structures due to the increase in space for circulation of the vortex. Introducing a pitching-plunging motion by offsetting the hind foil's pitching point further amplifies this constructive interaction and forms a greater, more refined vortex wake than pitching alone. These findings reveal that the optimal performance stems from leveraging specific spacing and kinematics to harness beneficial vortex interactions. These insights deepen the understanding of bio-inspired propulsion mechanisms and contribute toward the design of highly efficient propulsors for MAVs and AUVs.

Talk 4: Dr. Chandan Bose (*University of Birmingham*)

Unleashing GPU-Accelerated Computing: Aerodynamics of Butterfly Swarms

Chandan Bose, Debajyoti Kumar, Siddharth Sharma, Somnath Roy

The study of robotic butterfly swarms presents futuristic opportunities for understanding complex aerodynamic interactions in bio-inspired flight formations. However, simulating large-scale robotic swarms is computationally intensive due to intricate flow interactions, moving boundaries, and nonlinear phenomena. In this research, we harness the power of GPU-accelerated computing to accelerate aerodynamic simulations significantly using an Immersed Boundary Method and an overset meshing strategy. Our GPU-accelerated flow solver efficiently handles the complex flow interactions inherent to the flapping wings of robotic butterflies, enabling detailed characterisation of aerodynamic performance metrics such as lift, thrust, and propulsion efficiency. Through high-resolution computational fluid dynamics simulations, we explore critical aerodynamic features emerging within robotic butterfly swarms, including wake interactions, vortex formation, and aerodynamic synchronisation between multiple units. These insights not only deepen our fundamental understanding of swarm aerodynamics but also directly inform the design principles of next-generation robotic aerial systems. Ultimately, this study demonstrates the transformative potential of GPU computing in computational aerodynamics, offering robust and scalable tools capable of unlocking new paradigms in bio-inspired engineering and robotics research.

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